EXPERIMENT – 3

SURVEY REPORT ON THE EMERGING TRENDS IN LOW-CODE APPLICATION DEVELOPMENT

AIM

To study and analyze the emerging trends in low-code application development, including new tools, technologies, automation capabilities, artificial intelligence, and their advantages in modern software development.

INTRODUCTION

Low-code application development is a software development approach that uses visual interfaces, drag-and-drop components, pre-built modules, and minimal manual coding. It enables applications to be developed faster while reducing the technical effort required.

The adoption of low-code platforms is increasing because organizations require faster application development, process automation, and efficient utilization of development resources. Modern low-code platforms are also incorporating artificial intelligence, cloud services, automation, and advanced integration capabilities.

EMERGING TRENDS IN LOW-CODE APPLICATION DEVELOPMENT

1. AI-Assisted Development

Artificial Intelligence is increasingly being integrated into low-code platforms to generate application components, suggest workflows, assist with data modelling, and help users create applications using natural language instructions.

2. Intelligent Automation

Low-code platforms are combining application development with workflow automation. Repetitive business processes can be automated using triggers, workflows, schedules, and rule-based actions.

3. Integration with Cloud Services

Modern low-code platforms provide integration with cloud applications, databases, APIs, and third-party services. This allows applications to communicate with different systems without requiring extensive integration code.

4. Mobile Application Development

Low-code platforms increasingly support the development of responsive and mobile applications. Developers can create applications that work across different devices using common visual development tools.

5. Visual Data and Analytics

Advanced reporting, dashboards, data visualization, and analytics capabilities are becoming important features of low-code platforms. Users can analyze application data without developing complete analytics systems manually.

6. Increased Use of Reusable Components

Reusable components, templates, connectors, and modules allow developers to create applications faster and maintain consistency across multiple applications.

EMERGING LOW-CODE TOOLS AND PLATFORMS

1. Zoho Creator
   A low-code platform for creating business applications using forms, reports, dashboards, workflows, integrations, and Deluge scripting.

2. Microsoft Power Apps
   A Microsoft low-code platform used to build business applications and integrate them with Microsoft services and other data sources.
3. OutSystems
   A low-code application development platform designed for building and deploying enterprise applications.
4. Mendix
   A low-code platform that provides visual development, workflow automation, integration, and application lifecycle management capabilities.
5. Salesforce Platform
   Provides low-code tools for developing applications and automating business processes within the Salesforce ecosystem.

ADVANTAGES OF EMERGING LOW-CODE TECHNOLOGIES

- Faster application development.
- Reduced amount of manual coding.
- Lower development effort and cost.
- Rapid prototyping and testing.
- Easier workflow automation.
- Integration with external applications and cloud services.
- Improved collaboration between technical and non-technical users.
- Reusable components and templates.
- Support for data visualization and analytics.
- AI-assisted development and automation.

CONCLUSION

Low-code application development is evolving from simple visual application builders into intelligent development platforms that combine automation, artificial intelligence, cloud services, integration, analytics, and reusable components. These developments are helping organizations build and modify applications faster while reducing development effort.

The future of low-code development is expected to focus increasingly on AI-assisted development, intelligent automation, cloud integration, and simplified application development.

RESULT

Thus, a survey of the emerging trends in low-code application development was successfully conducted. Various modern trends, low-code platforms, tools, and advantages were studied, providing an understanding of how low-code technologies are evolving to support faster and more intelligent application development.